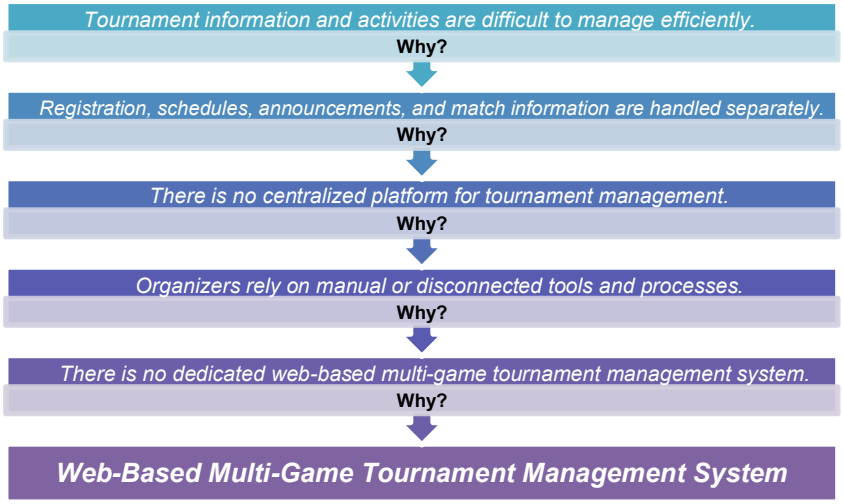
BUSINESS CASE	BSIT 31101

Project Title:	**DALTA Esports: Tournament Management System with AI-Powered Feedback Sentiment Analytics**
Client:	**DALTA Esports - UPHSD Molino, AcadArena South Luzon Esports Conference (SLEC) - Alliance Games**
Users:	**AcadArena SLEC - Alliance Games Organizing Committee (admin)**
	DALTA Esports Tournament Staff / Game Administrators (admin)
	Players and Spectators (end-users)

Project Team Members:	**Project Leader**	**Jhane Rose U. Sadicon**
	Systems Analyst	**Jhane Rose U. Sadicon**
	Developer	**Ma. Isabella B. Arce**
	Researcher/ Technical Writer	**Gianna Trisha M. Ainza**

BUSINESS CASE	BSIT 31101

<table>
<tr><td>Identified Problem:</td><td colspan="3">Tournament information and management are handled through separate/manual processes.</td></tr>
<tr><td>Root Cause Analysis:
(5 WHYs or Fishbone Analysis)</td><td colspan="3"> 
</td></tr>
<tr><td>Problem to be solved:
(Root cause)</td><td colspan="3">Web-Based Multi-Game Tournament Management System</td></tr>
<tr><td rowspan="4">Objectives:</td><td>General Objective:</td><td colspan="2">To develop a web-based multi-game tournament management system for the DALTA Esports and AcadArena Southern Luzon Esports Conference.</td></tr>
<tr><td rowspan="3">Specific Objectives:</td><td colspan="2">To develop a centralized module for player and team registration and tournament management.</td></tr>
<tr><td colspan="2">To provide event information, announcements, match updates, and live game streaming through the website.</td></tr>
<tr><td colspan="2">To collect and analyze participant and viewer feedback using sentiment analysis for committee decision-making.</td></tr>
<tr><td>Project Team Resources:
(Feasibility)</td><td>Human Resource:</td><td>Knowledge:</td><td>• Tournament management process
• Web development and maintenance</td></tr>
</table>

			• Database management • User feedback and sentiment analysis
		Skills:	• Web programming • Database design • UI/UX design • System deployment • Data analysis • Technical writing • Logical diagrams
	Technical Components:	**Front End: (Programming Language)**	HTML, CSS, JavaScript
		Back End: (Programming Language)	JavaScript (Node.js, Chart.js)
		Back End: (Database Application)	Supabase
		Sentiment Analysis	Gemini API, Python
		Deployment Platform	Railway
		Version Control	Git / GitHub
		Live Streaming	Facebook Live / Streaming API
	Budget (if applicable)	**Proposed Budget:**	Minimal budget of ₱0–₱5,000; primarily uses free and open-source software and existing hardware.
System Features:	**Account Management Module:** this module will allow users to create accounts and provide role-based access to system features.		

	Registration and Tournament Management Module: this module allows players and teams to register and enables administrators to manage tournament participants.
	Event Information and Announcements Module: this module provides tournament schedules, mechanics, announcements, and match updates.
	Live Streaming Module: this module provides access to live tournament game streams through the website.
	Feedback and Sentiment Analysis Module: this module will collect user feedback and analyze sentiments to identify common concerns and event insights.
	Administrative Reports Module: will generate summaries and reports from registration, tournament, and feedback data.
	Database Module: will store and manage player, team, tournament, match, and feedback records.